
Enhanced GIS-Based Spatial Correlation Analysis of Industrial Areas and Urban Drainage Using Hydrological Flow Connectivity and Water-Quality Sensor Data: A Case Study of Tandi Ditch, Dahan River Watershed

1. INTRODUCTION

Taiwan's rivers are constantly under pollution pressure from the factories and the high density of living people around them. Pollution does not appear naturally in a river; it starts from the ground and travels through the urban drainage system (ditches, culverts, and storm channels) and then flows into the main river. Drainage is therefore the critical link between the land-based sources and the receiving river. Nevertheless, most environmental monitoring measures water quality in the river after the pollution has arrived, as the River Pollution Index indicates and is used by such indices. This means that a river is polluted but it is not yet known where the load came from and how it happened or who is responsible. The land-to-drainage relationship that informs this is rarely mapped at a local level and city scale. This study's guiding principle is that an industrial source is genuinely susceptible to contaminating the river only when three independent conditions coincide at the same place: (i) the source is close to drainage; (ii) the terrain routes runoff toward that drainage; and (iii) measurements show the water there is already under stress. Each condition is necessary; none alone is sufficient. We capture the three conditions plus the relative polluting potential of different industries in four GIS layers and combine them into a single Pollution Susceptibility Index (PSI) through weighted overlay. The integration of real water-quality sensor data is the decisive addition, because it is what lets the analysis move from "things are close" to "these specific sites combine a real pathway with measured degradation."

Objectives. (1) Measure proximity of industrial areas to drainage using buffer and nearest-neighbor analysis; (2) model hydrological flow connectivity from a DEM so that distance follows real drainage direction; (3) integrate a water-quality stress index from in-channel sensor stations; (4) weight industrial sources by type and area; (5) combine these in a weighted-overlay PSI and validate it statistically against the sensor data; and (6) classify, map, and rank High / Moderate / Low susceptibility sites to guide targeted monitoring and enforcement.

2. STUDY AREA

The study area is the industrial area of Xinzhuang District, New Taipei City, and its associated drainage. Xinzhuang is part of the Dahan River watershed, one of the main branches of the Tamsui River system that drains the Taipei Basin. The district is very dense and has many factories and homes in it, so many potential pollution sources are situated close to the ditches and culverts that carry runoff to the Dahan River. Sixty-

five industrial areas were investigated. The water quality is determined by the Ministry of Environment's river monitoring network on the Dahan River. Seven stations were analyzed to measure the upstream to downstream gradient, and the two bridges that converge on Xinzhuang (Fuzhou Bridge (浮洲橋) and Xinhai Bridge (新海大橋) were assigned to each industrial area by nearest-station join.

Study area: drainage network, industrial areas, and the Dahan River and Water-Quality Monitoring Stations in Xinzhuang District, New Taipei City

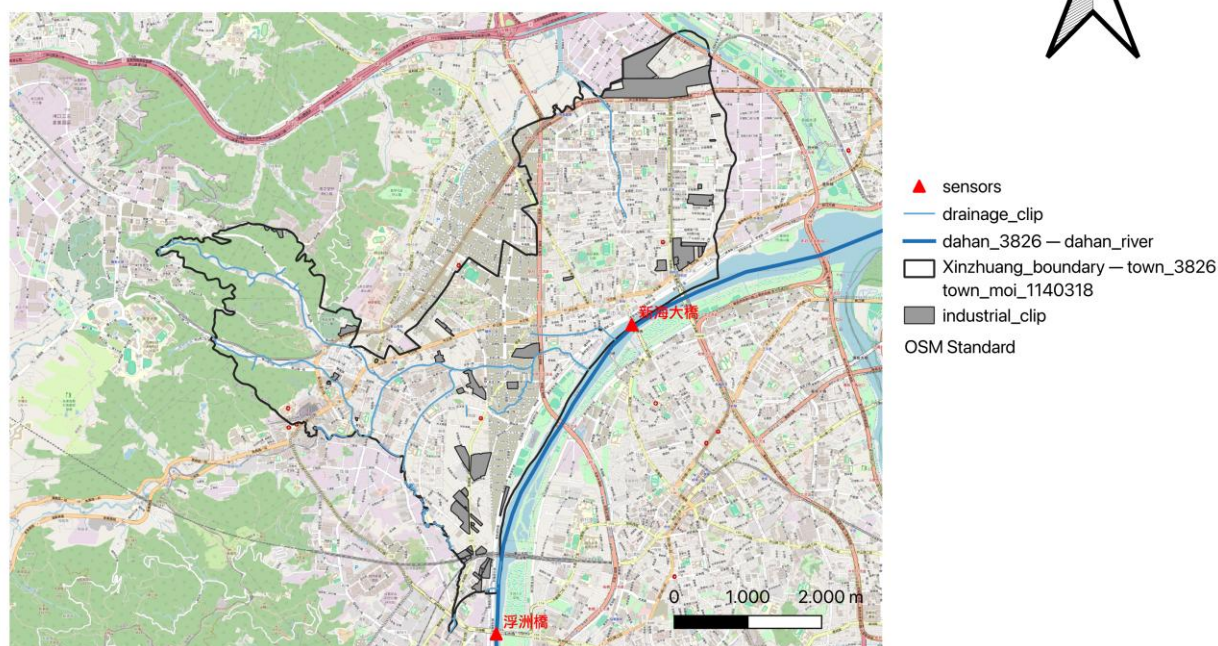


Figure 1. Study area: drainage network, 65 industrial areas, the Dahan River, and the MoENV water-quality stations within Xinzhuang District.

3. DATA COLLECTION

Five categories of spatial data were collected. OpenStreetMap provided drainage and industrial layers; official Taiwanese open data provided the district boundary and terrain; and the Ministry of Environment provided the measured water-quality readings. The water-quality data were not entered manually, but the two monthly MoENV open-data export files were combined and cleaned in Python (pandas) to build a reproducible station table.

Table 1. Datasets used in this study.

Dataset	Source (link)	Role in analysis
Drainage network	OpenStreetMap (waterway/ditch/culvert/drain) (https://www.openstreetmap.org)	Pathway - buffered + flow
Industrial areas (65)	OpenStreetMap (landuse=industrial) (https://www.openstreetmap.org)	Source units
DEM (terrain)	NLSC Taiwan DTM 20 m (https://portal.opentopography.org)	Flow / connectivity
Water-quality stations	MoENV river water-quality open data - 7 Dahan River stations (combined from two CSV files via Python: https://data.moenv.gov.tw/en/dataset/detail/WQX_P_01 ; https://data.moenv.gov.tw/en/dataset/detail/WQX_P_06)	Measured-condition layer
Boundary & river	NLSC township SHP; river from OSM (https://data.gov.tw/en/datasets/7441)	Context & receiving river

Table 2. Measured water quality at the seven MoENV Dahan River stations

Station	DO (mg/L)	Cond. (μ S/cm)	NH ₃ -N (mg/L)	pH
Houchi Weir (後池堰)	8.6	234	0.03	7.96
Daxi Bridge (大溪橋)	8.9	286	0.13	7.78
Banxin Intake (板新取水口)	7.7	204	0.22	7.78
Sanying Bridge (三鶯大橋)	8.1	220	0.15	7.74
Ganyuan 2nd Bridge (柑園二橋)	7.4	306	0.63	7.70
Fuzhou Bridge (浮洲橋)	6.5	238	0.78	7.55
Xinhai Bridge (新海大橋)	5.8	222	0.99	7.41

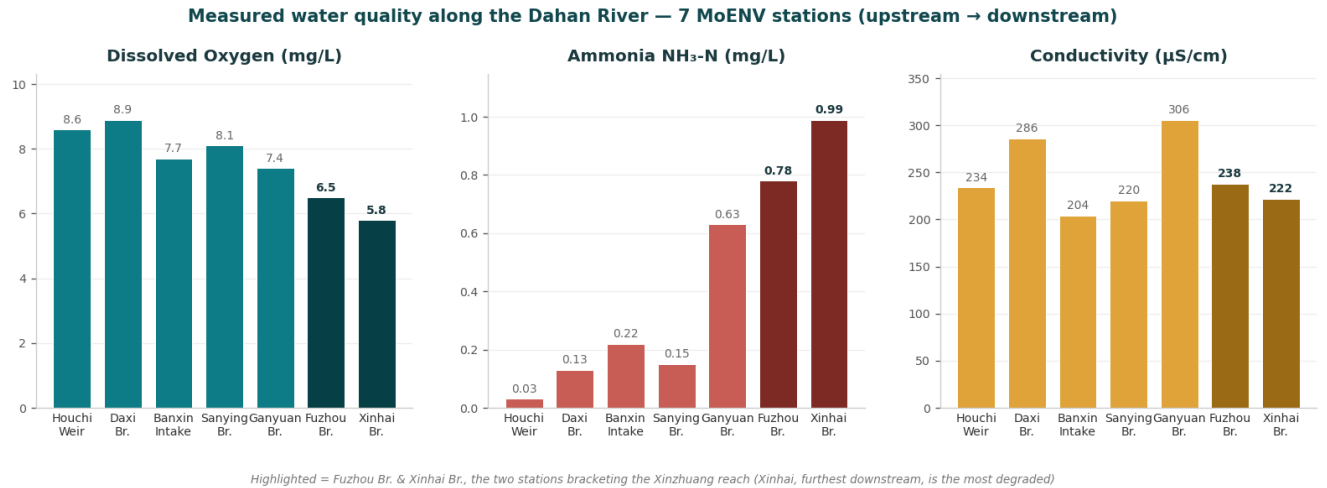


Figure 2. Measured water quality along the Dahan River. Dissolved oxygen falls and ammonia rises toward the downstream stations; Xinhai Bridge (the most downstream, nearest the Xin Zhuang outfall) shows the lowest oxygen and highest ammonia, indicating the greatest pollution load.

4. METHODS

The workflow consists of six steps in QGIS (Figure 3). Four evidence layers (distance, flow connectivity, sensor water quality, and source intensity) are normalized to the 0–1 score and weighted together to form the PSI.

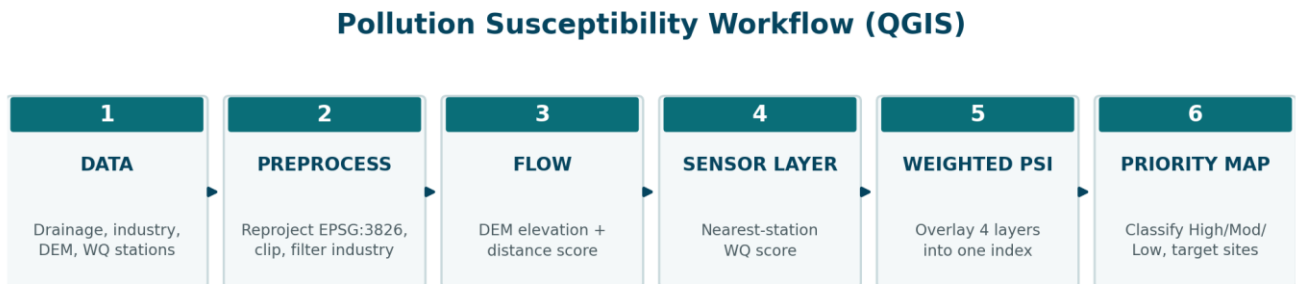


Figure 3. Six-step GIS workflow: Data → Preprocess → Flow connectivity (DEM) → Sensor layer → Weighted PSI overlay → Priority map.

4.1 Layer 1 - Proximity

Buffers of 100 m, 300 m and 500 m were created around the drainage network. A near-neighbour spatial join took the distance of each industrial area to the nearest channel and rescaled it to a proximity score with closer = higher: $P = 1 - (d / 1200)$.

4.2 Layer 2 - Flow connectivity (DEM)

The ground elevation at each industrial area was measured with the Zonal Statistics tool (mean elevation). The connectivity score was then computed within the QGIS Field Calculator without the GRASS or SAGA model: the normalized elevation of each parcel was inverted and multiplied by its normalized proximity, so parcels which are lower-lying and closer to the channel (the ones with the highest likelihood of reaching the ditch) were scored highest. The hydrological connectivity was therefore determined with a slope-and-distance proxy from the DEM instead of a full flow accumulation model, an obvious simplification to deal with the data.

4.3 Layer 3 - Water-quality sensors

The water-quality layer was created directly from the Ministry of Environment's river water-quality monitoring open data from different sources (and not from values keyed in by hand). The two monthly export files were concatenated in Python (pandas) and filtered to the records with the Dahan River in the river field (大漢溪). The researchers searched for pH (酸鹼值), conductivity (導電度), dissolved oxygen by electrode method (溶氧 (電極法)), ammonia-nitrogen (氨氮) and turbidity (濁度). For each monitoring station, the most recent record of each parameter was kept, and the data was fed into a station-by-parameter table, and the coordinates (TWD97) were added to the MoENV station information file (the station coordinates were dropped if there was no coordinates). This created a clean, reproducible table for the real measurements of the values for seven Dahan river stations (Table 2). The turbidity was not found for the latest sampling dates at these stations so the stress score is based on dissolved oxygen and ammonia. Each industrial area was then assigned the readings of its closest station by spatial join and given a 0-1 stress score. The two stations in the Xinzhuang reach drive this layer: the score tells us the degraded downstream zone (Xinhai Bridge: DO 5.8 mg/L, NH₃ 0.99 mg/L) from the clean upstream zone (Fuzhou Bridge: DO 6.5 mg/L, NH₃ 0.78 mg/L) and thus the quality of the river has fallen from around 8-9 mg/L in the upper reach down to 5.8 mg/L. This is the sensor data integration (感測器數據整合).

4.4 Layer 4 - Source intensity

Each industrial polygon got a source intensity weight. In our case, the OSM industrial polygons did not have a reliable sub-type attribute, so we used a uniform weight (0.6) for all the areas, and this is a relatively conservative mid-range value for the light-industry tier of the proposed scheme. The model is laid out so that when sub-type data are available, heavy industry can be up weighted with respect to warehousing without any other step.

4.5 Weighted overlay - the PSI

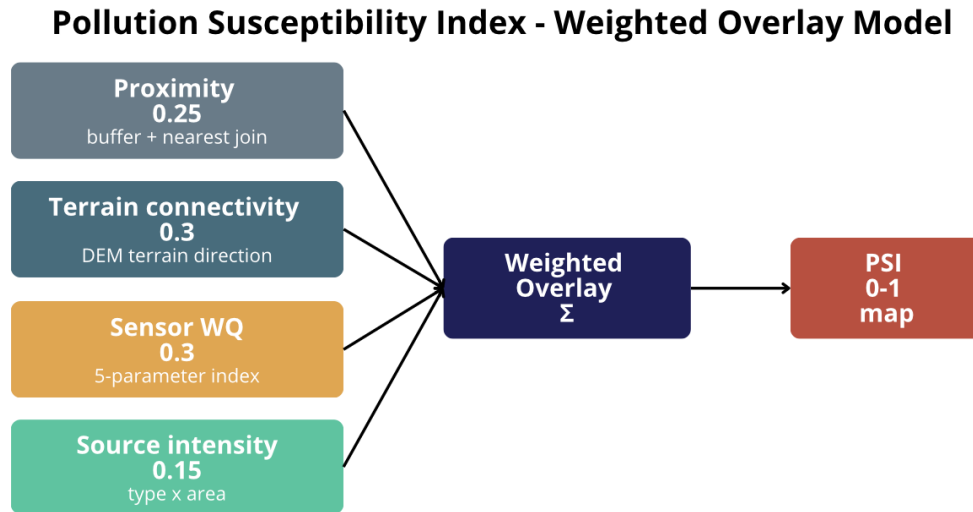


Figure 4. Weighted-overlay model: The four normalized layers were combined to produce a 0-1 Pollution Susceptibility Index.

The four layers were combined with the field calculator:

$$PSI = (0.25 \times \text{Proximity}) + (0.30 \times \text{Flow connectivity}) + (0.30 \times \text{Sensor WQ}) + (0.15 \times \text{Source intensity}).$$

The flow connectivity and sensor evidence have a lot more weight (0.30 each) because a short distance matters. We divided the PSI into three classes by natural breaks (Jenks). The cut points of the PSI were 0.614 and 0.737.

5. RESULTS AND DISCUSSIONS

5.1 Susceptibility classification

The PSI ranges from 0.354 to 0.936 for the 65 industrial areas (average 0.622). The natural breaks classification gives 12 High (18%), 17 Moderate (26%), and 36 Low (55%) areas (Table 3, Figure 5). The High class is clearly distinguished as it is only 76 m away from the drainage and has good connectivity (0.91) and the highest sensor-stress score (1.00) compared to the Low, which is further (431 m) and less connected (0.60).

Table 3. Industrial areas by Pollution Susceptibility class, with mean layer values.

Class	n	%	Mean dist. (m)	Mean conn.	Mean PSI
High (PSI ≥ 0.737)	12	18.5	76	0.91	0.898
Moderate (0.614–0.737)	17	26.2	102	0.89	0.639
Low (PSI < 0.614)	36	55.4	431	0.60	0.523

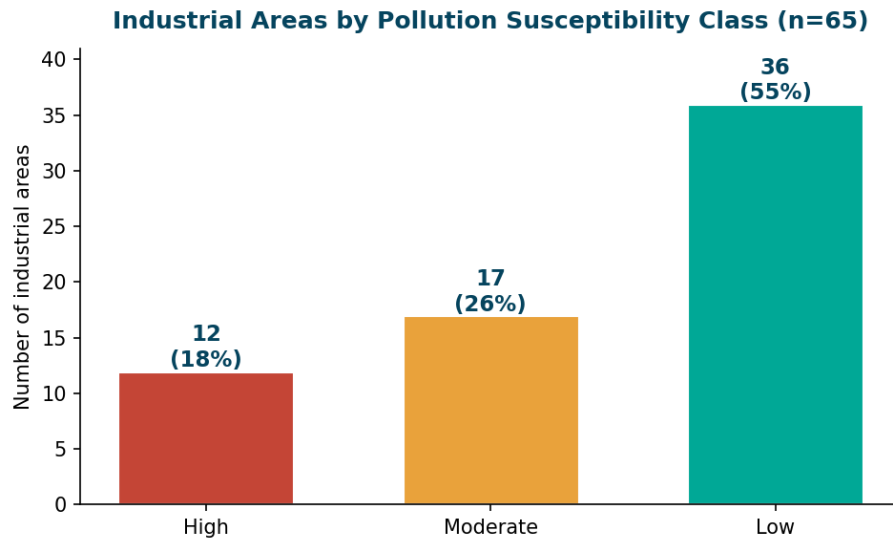


Figure 5. Distribution of the 65 industrial areas across the three susceptibility classes.

5.2 Each layer contributes to the index

All four evidence layers contribute to the final score instead of one dominating. Proximity, flow connectivity and sensor score all correlate positively and similarly with the PSI ($r = 0.52$, 0.51 and 0.49 respectively in Figure). Proximity and flow connectivity are similarly strongly related ($r = 0.96$) and this is not surprising since parcels near a ditch also tend to be low-lying. The balance of correlations tells us that a site is in the High class when it is close, drained towards the channel and is in the water-quality zone rather than just distance.

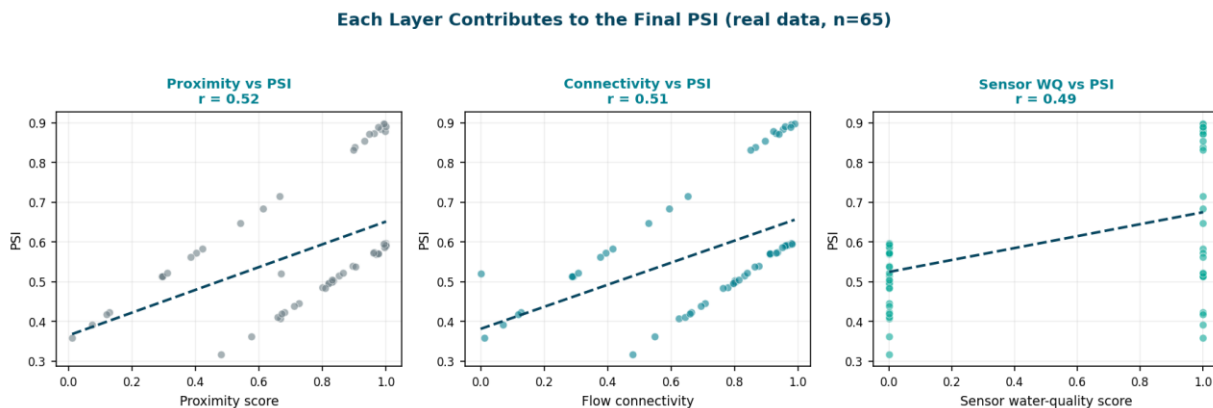


Figure 6. Each evidence layer plotted on top of the final PSI. All three contribute positively ($r = 0.49$ - 0.52) showing the index is a combination of factors.

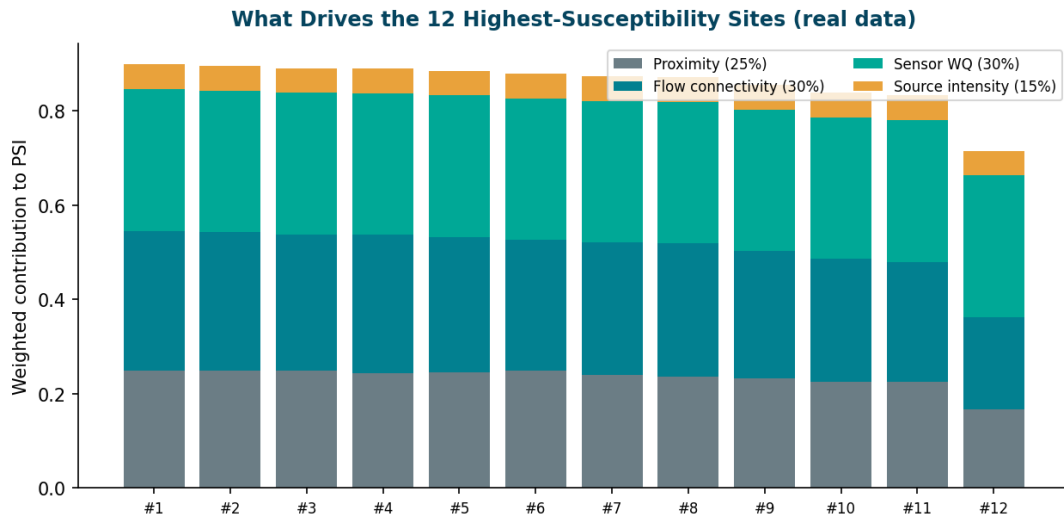


Figure 7. Contribution of each layer to the 12 highest-susceptibility sites. Proximity, flow connectivity and (downstream) sensor score all add significantly.

5.3 Priority sites cluster in space

The High-susceptibility areas are not scattered randomly. Average nearest neighbor analysis provides $R = 0.51$ ($z = -3.24$, $p < 0.01$), which means they are highly clustered and not dispersed (Figure 8). Operationally this matters: a cluster of priority sites can only be addressed by a small number of targeted approaches such as a concentrated inspection sweep or a cluster of new sensors but not the whole cluster over all 65 zones. The downstream zone served by the Xinhai Bridge station which has the poorer measured water quality also has a higher mean PSI (0.71 versus 0.56 in the upstream zone) which links the spatial pattern to the pollution gradient.

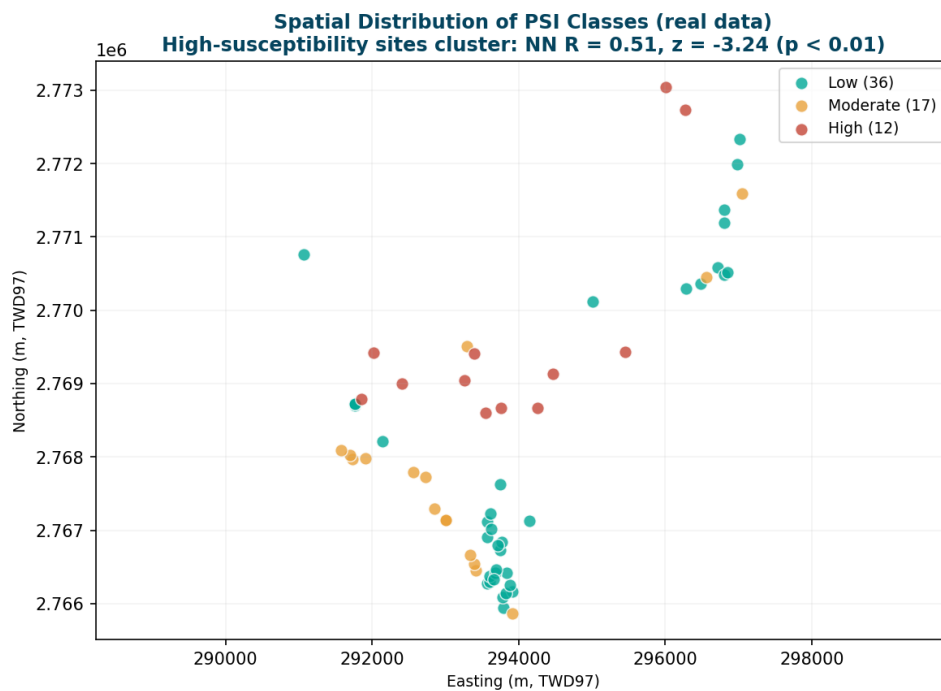


Figure 8. Spatial distribution of PSI classes. The high-susceptibility sites (red) form a statistically significant cluster (nearest-neighbor $R = 0.51$ and $z = -3.24$).

Pollution Susceptibility Index (PSI) priority map: industrial areas classified High, Moderate, and Low

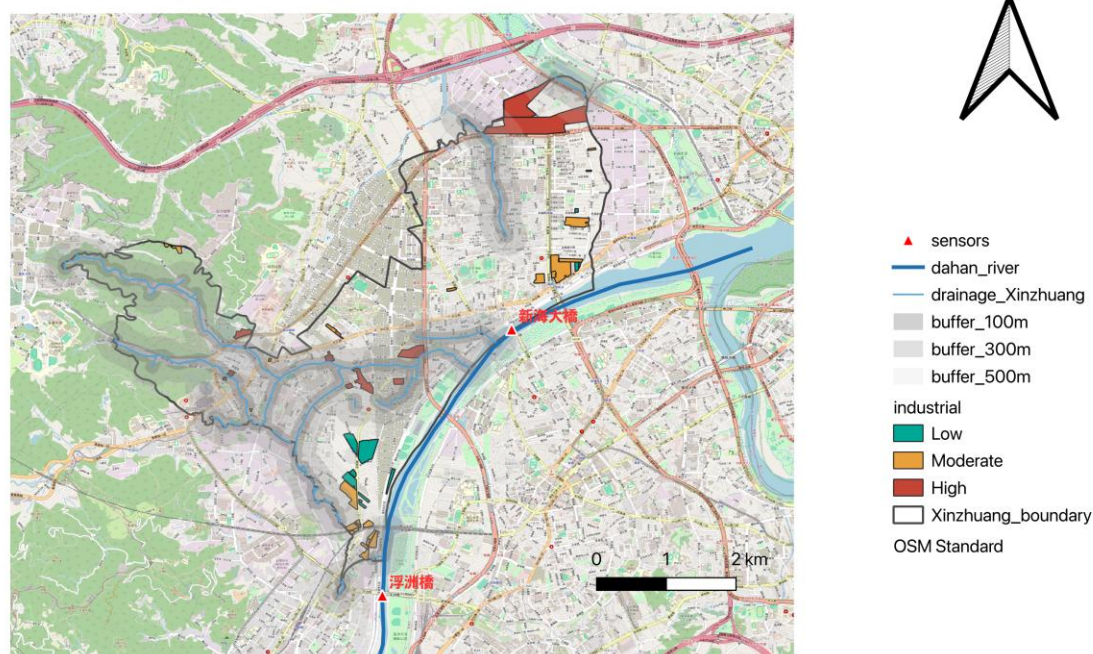


Figure 9. PSI priority map: industrial areas graduated High (red) / Moderate (amber) / Low (green) over the drainage network and sensor stations. (Produce in QGIS.).

5.4 What this analysis is for

The PSI answers the practical question a water authority faces: given a limited inspection budget, which sites should be checked first? The 12 High-class areas, concentrated in one spatial cluster, are a safe shortlist: each is close to drainage, drains towards the ditch, and falls into the more degraded downstream water quality area. The New Taipei City Water Resources Bureau or Ministry of Environment might direct inspectors and source-control enforcement to this cluster, and prioritize IoT sensor deployment around it, rather than treating all 65 areas equally. The model also takes an analytical step earlier than traditional monitoring: instead of waiting to measure pollution in the river, it looks at sources and pathways in place and what the stations already detect.

5.5 Contribution to the SDGs

- SDG 6 (Clean Water and Sanitation): identifies where pollution most likely enters the drainage-to-river pathway, which makes it possible to have water quality targeted to the area.
- SDG 11 (Sustainable Cities and Communities): provides planners with a reusable and evidence-based approach for the industry-drainage interface in dense urban areas.

- SDG 13 (Climate Action): the flow-and-sensor framework expands to runoff and storm-load monitoring as rainfall patterns increase.

6. CONCLUSION

In this study, we developed a multi-criteria Pollution Susceptibility Index for 65 industrial areas of Xinzhuang. Our analysis combines buffer/proximity, DEM-based flow connectivity with MoENV water quality data and source intensity with weighted overlay in QGIS. We identify 12 areas (18%) as High susceptibility areas, which are a statistically significant cluster (nearest-neighbor $R = 0.51$) with the more degraded downstream water quality zone. These are linked to the quality of the water by a balance of proximity, connectivity and measured water quality (with each region being $r \approx 0.5$ on the index). From “things are close” to “these particular sites are well placed to provide a real pathway together with measured degradation”, the data on these areas in the study yields a good, high priority list to work with field inspection and sensor deployment. Flow connectivity is a DEM-based slope-and-distance proxy, rather than a full flow-accumulation model. The source intensity was applied uniformly because reliable industrial sub-type data were not available. The future work should include more water-quality stations, move to a full flow model (instead of a proxy), add industrial sub-types to enhance intensity, validate with samples from the grab-sample laboratory and publish the priority map in a public web-GIS dashboard.

7. REFERENCES

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